

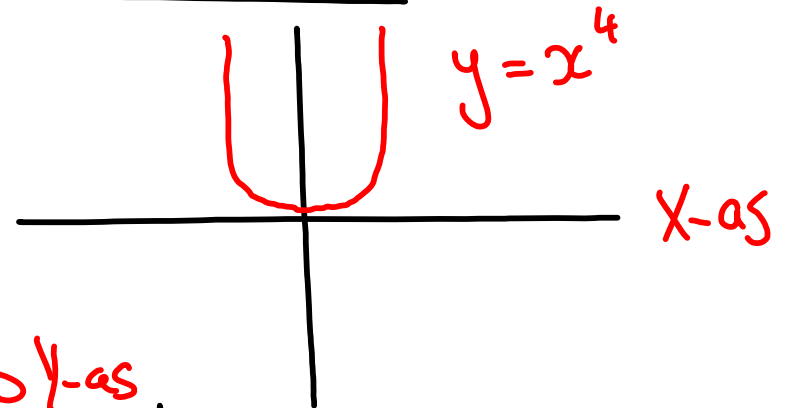
Nuwe funksies van ou funksies

Opsomming:

1. Verskuiwing van funksies
2. Refleksie van funksies
3. Uitrekking en kompressie
4. Kombinasies van funksies
5. Saamgestelde funksies

① Verskuiwing van funksies

Laat $y = x^4$

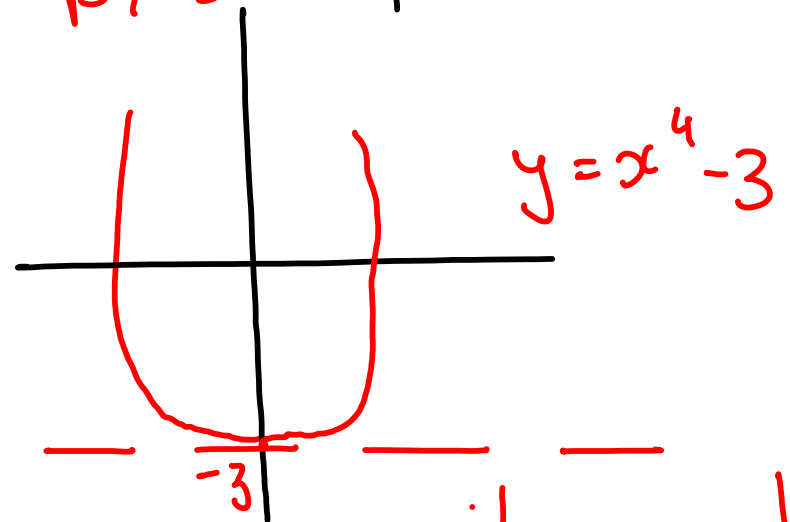


Vertikale skuif:

$$f(x) = x^4 - 3$$

A red arrow points from the '-3' in the equation to the graph below.

inloed op y-as



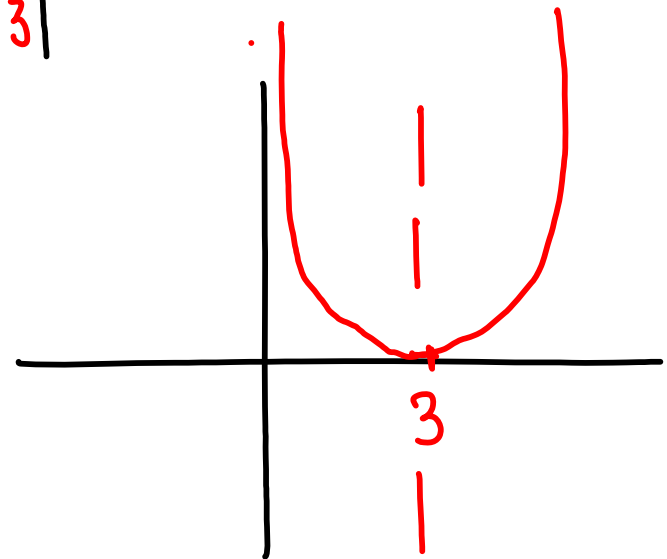
Horisontale skuif:

$$g(x) = (x - 3)^4$$

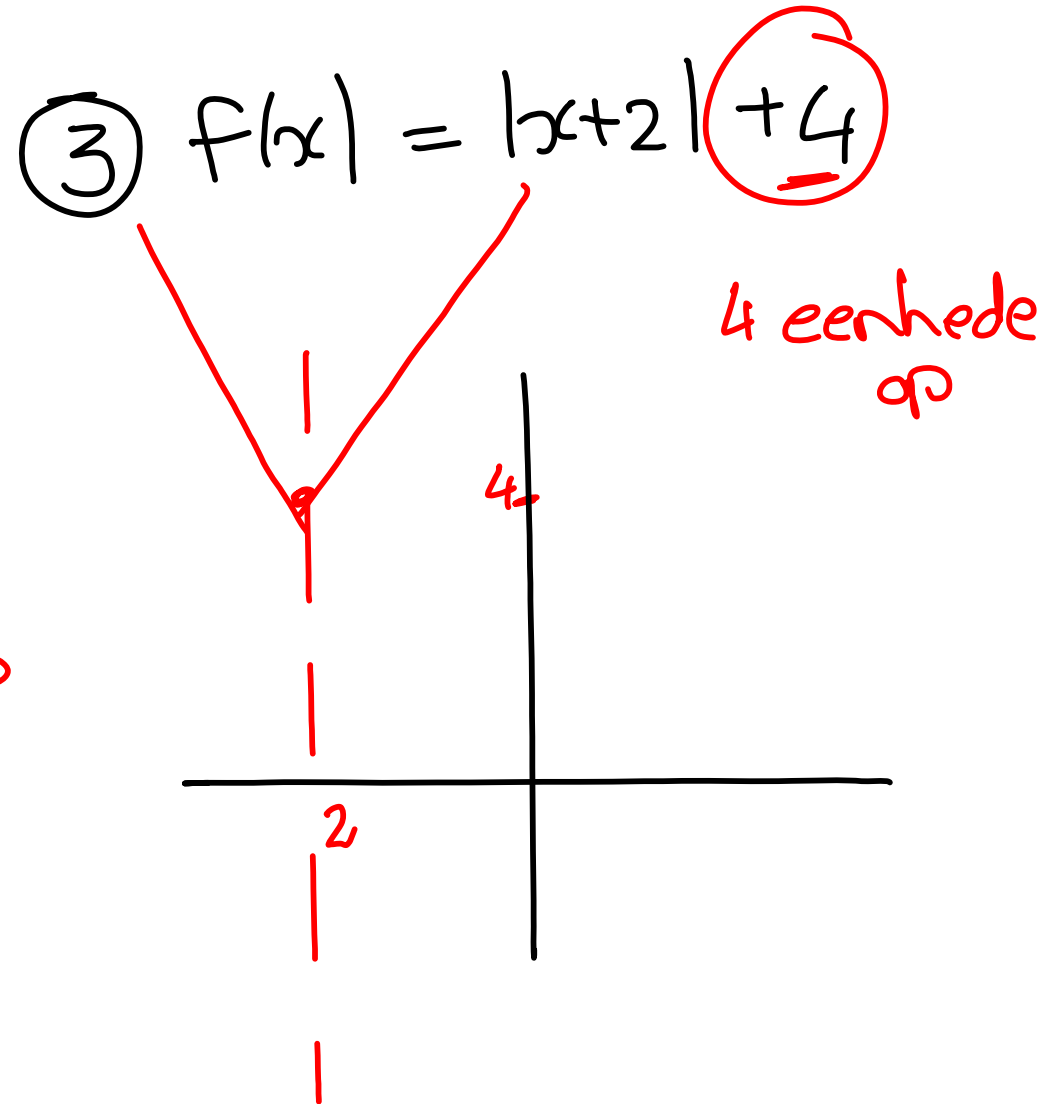
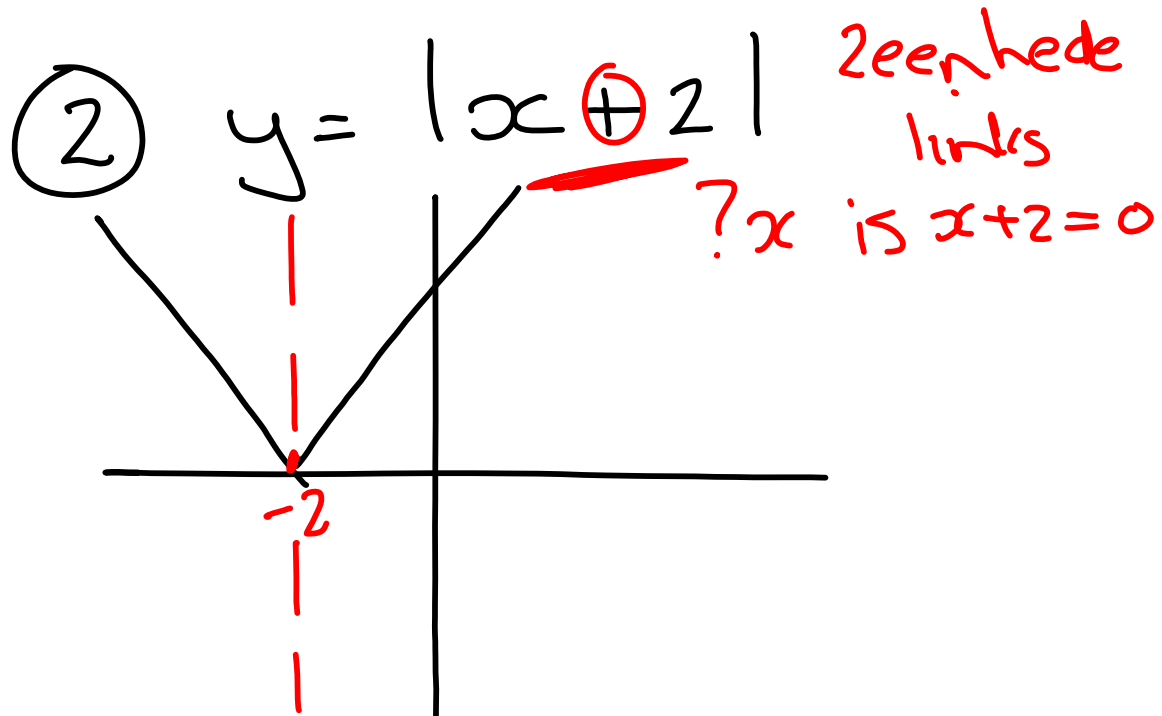
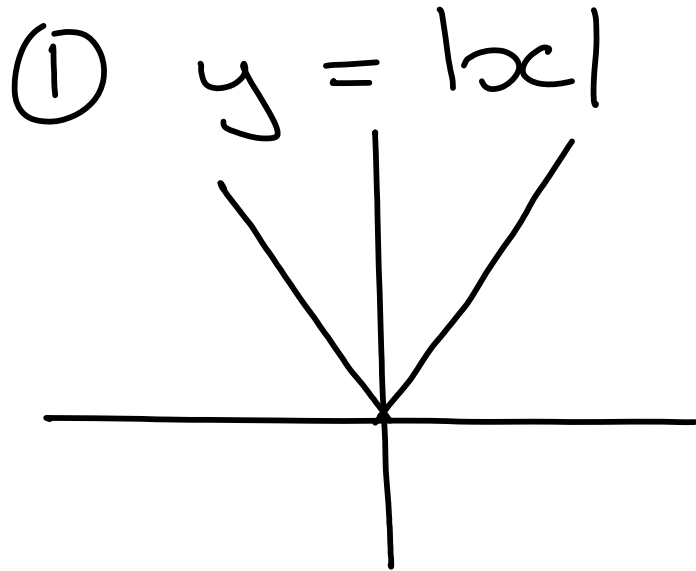
inloed op x-as

⊖ na regs

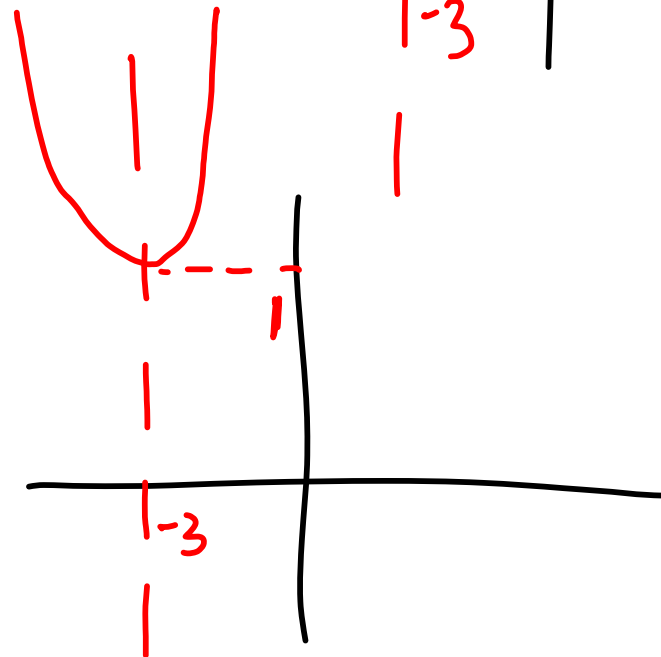
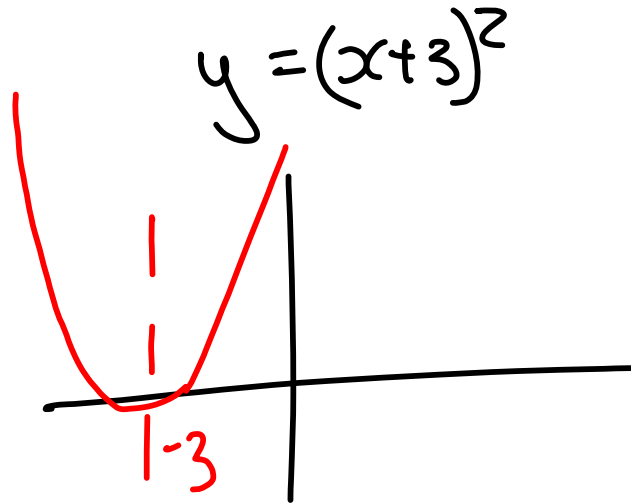
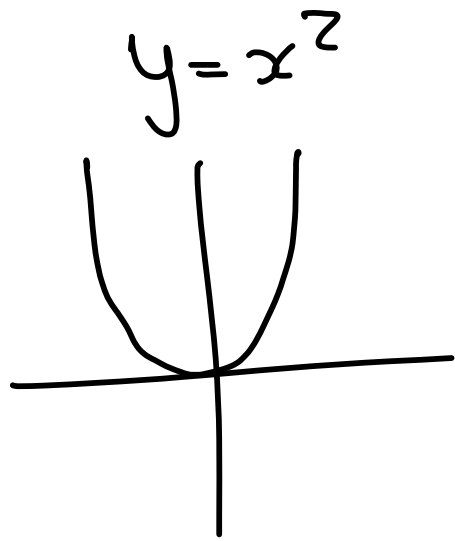
⊕ na links



Skets $\bullet f(x) = |x+2| + 4$



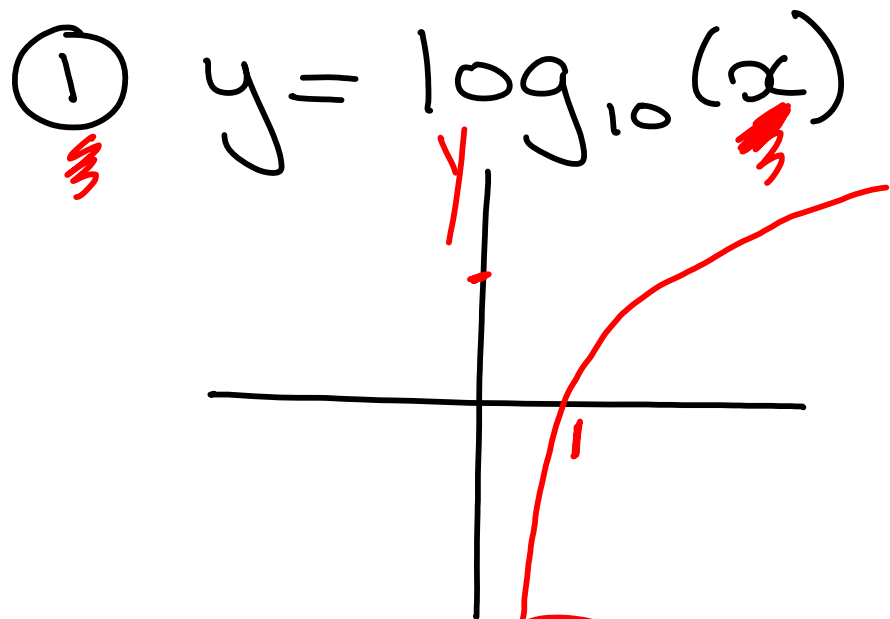
Sketch $f(x) = x^2 + 6x + 10$
 (Vierkantsvoltooiing) $= (x^2 + 6x + 9) - 9 + 10$
 $= (x+3)^2 + 1$



$y = (x+3)^2 + 1$

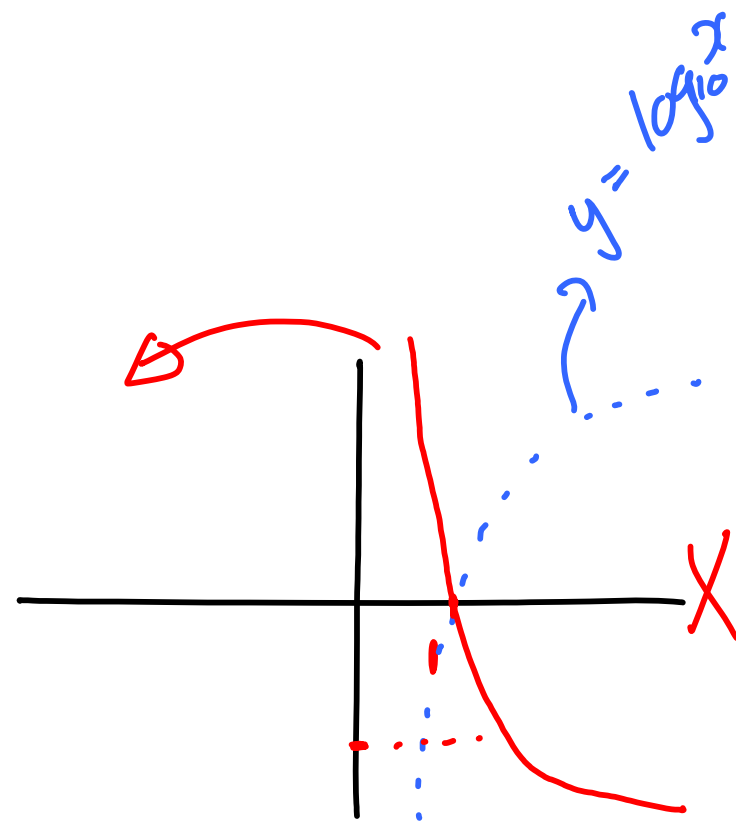
② Refleksie van funksies.

Skets $f(x) = -\log_{10}(x)$
en $g(x) = \log_{10}(-x)$



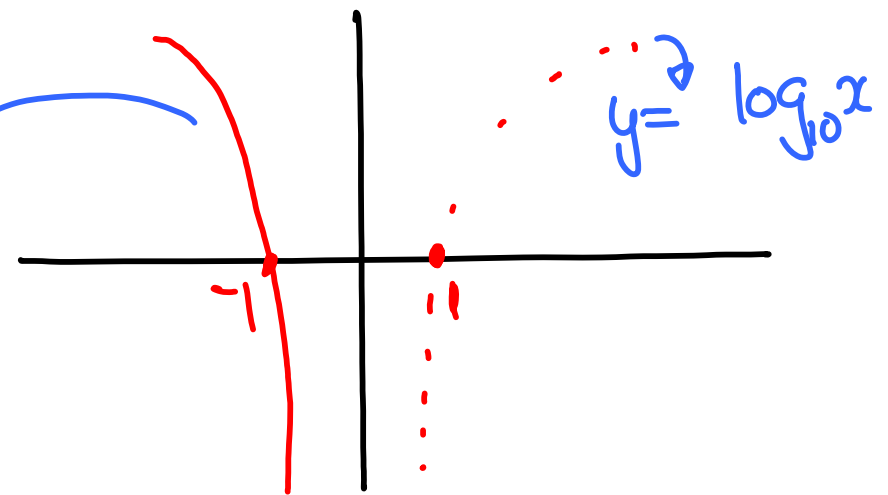
* ② $y = -\log_{10}(x)$

Reflekteer om x-as



③ $y = \log_{10}(-x)$

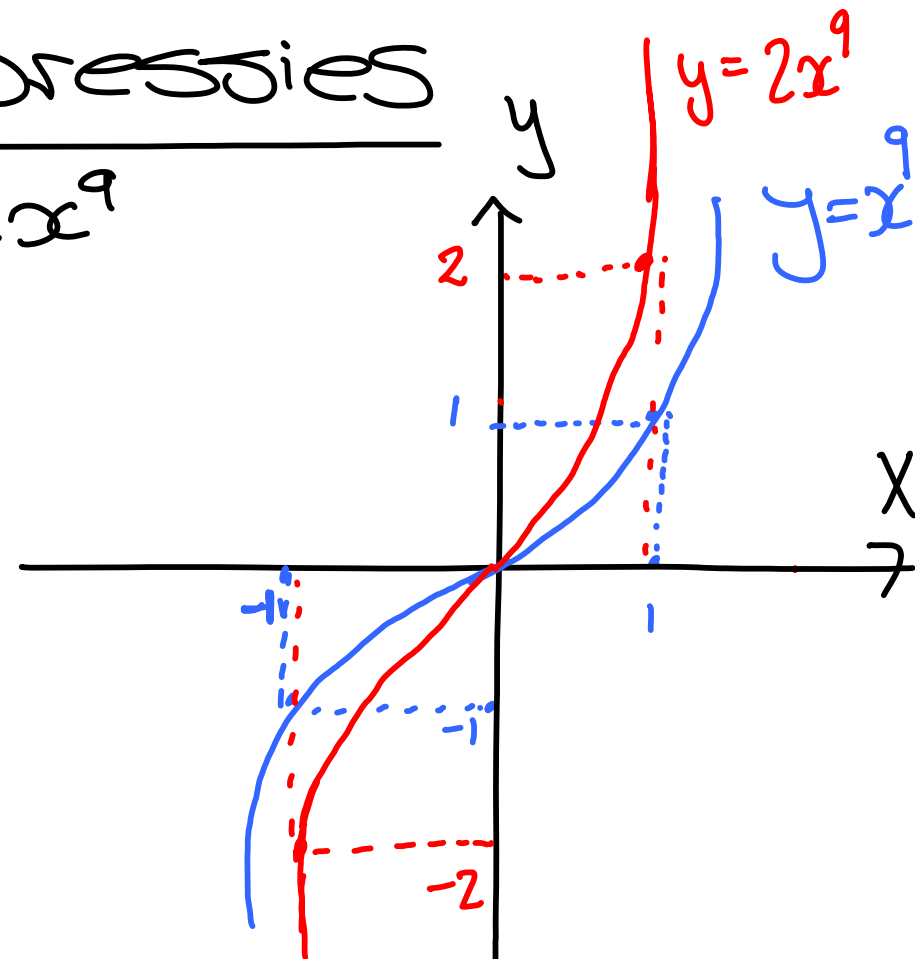
Reflekteer an y-as



③ Uitrekking en kompressies

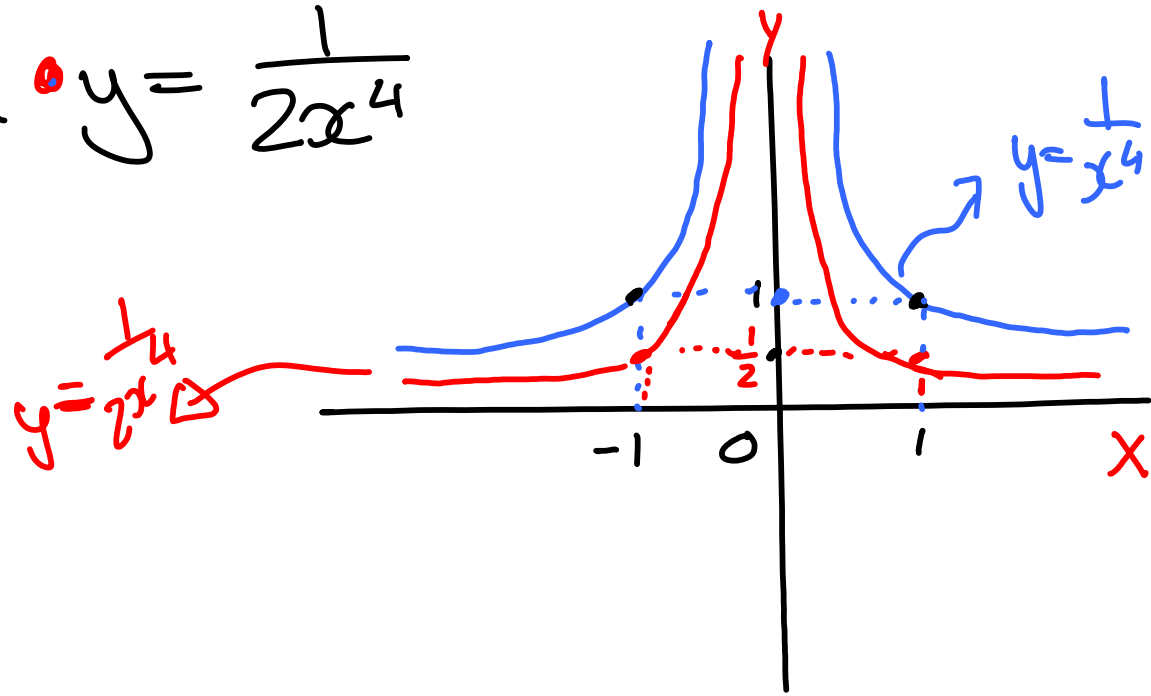
Skets $y = x^9$ en $y = 2x^9$

op die selfde
assessetel.



Skets $y = \frac{1}{x^4}$ en $y = \frac{1}{2x^4}$

op die selfde
as gestel



~~*Kyk na Trigonometrie funksies~~
Bv. $y = 1 \sin \theta$, $\theta \in [0, 2\pi]$

periode = 2π

amplitude = 1

Sien Eenheid 1.5 selfstudie

④ Kombinaties van functies p40

$$f(x) = \sqrt{x}, \quad g(x) = x^2$$

Bepaling van
Gebied is NB

$$(f+g)x = \sqrt{x} + x^2$$

$$D_f = [0, \infty)$$

$$D_g = \mathbb{R}$$

$$D_{f+g} = [0, \infty)$$

$$\bullet (f-g)x = \sqrt{x} - x^2$$

$$D_{f-g} = [0, \infty)$$

$$\bullet (fg)(x) = (\sqrt{x})(x^2)$$

$$D_{fg} = [0, \infty)$$

$$\bullet \left(\frac{f}{g}\right)(x) = \frac{\sqrt{x}}{x^2}$$

$$\text{Maar: } D_{\frac{f}{g}} = (0, \infty)$$

Tegriekbemeeste-
ring:

$$\bullet f(x) = \frac{\sqrt{x+1}}{\sqrt{x}-1} \quad D_f = ?$$

$$\sqrt{x} \geq 0 \Rightarrow x \geq 0 \quad \mathbb{N}$$

$$\sqrt{x} - 1 \neq 0 \Rightarrow x \neq 1$$

$$D_f = [0, 1) \cup (1, \infty)$$

⑤ Saamgestelde Funksies p41

Laat $g: A \rightarrow B$ en $f: B \rightarrow C$
twee funksies wees.

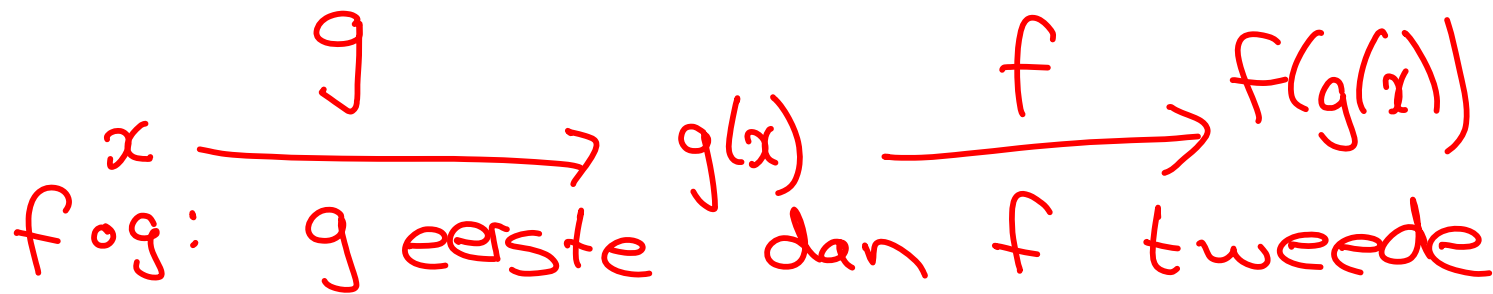
Dan word die ~~saamgestelde~~ funksie
 $f \circ g: A \rightarrow C$ gedefinieer deur

$$(f \circ g)(x) = f(g(x))$$

buite

binne

Begin binne



Voorbeelde

$$\textcircled{1} \ f(\underline{x}) = \underline{x}^2 \text{ en } g(\underline{x}) = \underline{x} + 1$$

$$\begin{aligned} \bullet \ (f \circ g)(x) &= f(g(x)) = f(x+1) \\ &= (x+1)^2 \\ &= x^2 + 2x + 1 \end{aligned}$$

$$\begin{aligned} \bullet \ (g \circ f)(x) &= g(f(x)) = g(x^2) \\ &= x^2 + 1 \end{aligned}$$

Dus $f \circ g \neq g \circ f$

$$\textcircled{2} \quad f(x) = 1 - 3x \quad \text{er} \quad g(x) = \cos x$$

$$\begin{aligned} \bullet (f \circ g)(x) &= f(g(x)) = f(\cos x) \\ &= 1 - 3 \cos x \end{aligned}$$

$$\begin{aligned} \bullet (g \circ f)(x) &= g(f(x)) = g(1 - 3x) \\ &= \cos(1 - 3x) \end{aligned}$$

③ Bepaal funkties f en g zodat

$$h = f \circ g \quad \text{waar} \quad h(x) = \sin x^2$$

$$\checkmark g(x) = x^2$$

$$f(\underline{x}) = \sin \underline{x}$$

$$\begin{aligned} \underline{\underline{\text{Toets:}}} \quad f(g(x)) &= f(\underline{x^2}) \\ &= \sin x^2 \end{aligned}$$

$$\begin{aligned} \textcircled{4} \quad h(x) &= \sin^2 x = (f \circ g)(x) \\ &= (\underline{\sin x})^2 \end{aligned}$$

$$g(x) = \sin x$$

$$f(\underline{x}) = (\underline{x})^2$$

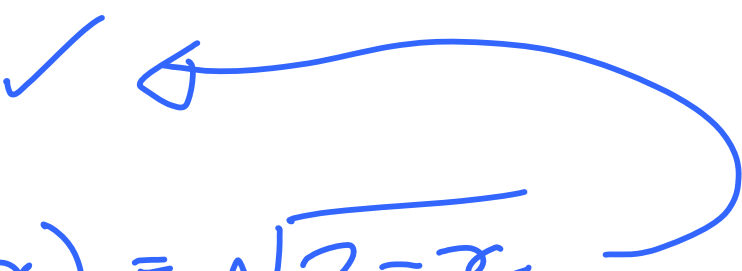
$$\underline{\underline{\text{Toets:}}} \quad f(\sin x) = (\sin x)^2$$

$$\textcircled{5} \quad h(x) = \sqrt{2-x} = (f \circ g)(x)$$

$$f(x) = \sqrt{x}$$

$$g(x) = 2-x$$

Tests: $f(2-x) = \sqrt{2-x}$



$$\textcircled{6} \quad (f \circ g)(x) = h(x) = \frac{1}{x-3}$$

$$g(x) = x-3$$

$$\textcolor{red}{f}(x) = \frac{1}{x}$$

Tests: $f(x-3) = \frac{1}{x-3}$

⑦ $F = f \circ g \circ h$ en $F(x) = \sqrt{\tan(3x)}$

$$h(x) = 3x$$

$$g(x) = \tan[x]$$

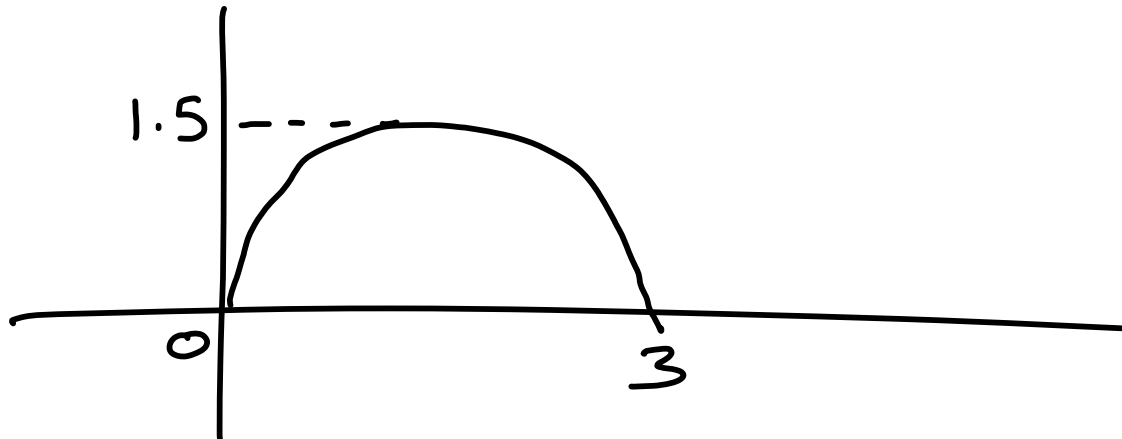


$$f(x) = \sqrt{x}$$

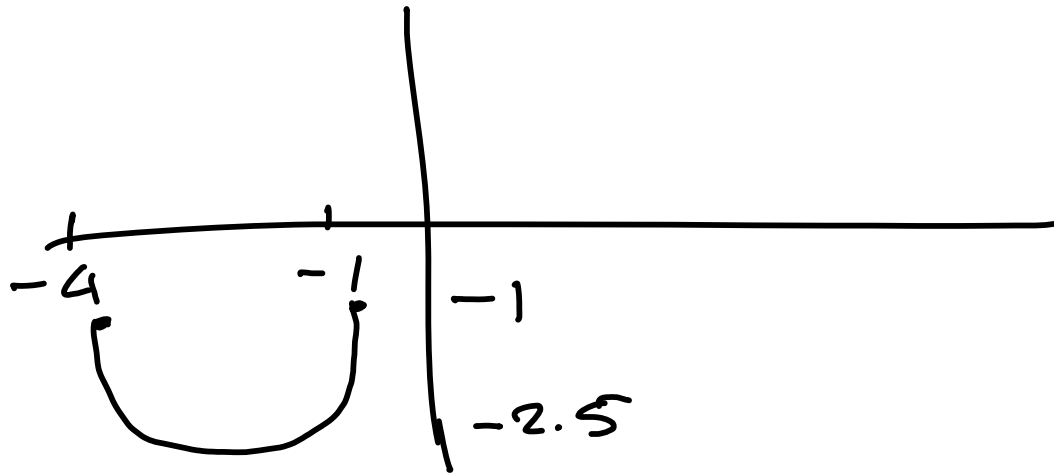
Stewart # ⑦ p43

$$y = \sqrt{3x - x^2}$$

(gegeve)



Hoe kom ek uit by hierdie grafiek?



22 Skets $y = 1 - 2\sqrt{x+3}$ P43

37

$$f(x) = x + \frac{1}{x}$$

$$D_f = \{x \mid x \neq 0\}$$

$$g(x) = \frac{x+1}{x+2}$$

$$D_g = \{x \mid x \neq -2\}$$

$$(f \circ g)(x) = f\left(\frac{x+1}{x+2}\right)$$

$$= \frac{x+1}{x+2} + \frac{1}{\frac{x+1}{x+2}} = \frac{x+1}{x+2} + \frac{x+2}{x+1}$$

$$x \neq -2, x \neq -1$$

$$= \frac{(x+1)(x+1) + (x+2)^2}{(x+2)(x+1)}$$

$$= \frac{2x^2 + 6x + 5}{(x+2)(x+1)}$$

$$D_{f \circ g}$$

$$= \{x \mid x \neq -2 \text{ en } x \neq -1\}$$

$$(g \circ f)(x) = g\left(x + \frac{1}{x}\right)$$

$$= \frac{\left(x + \frac{1}{x}\right) + 1}{\left(x + \frac{1}{x}\right) + 2}$$

$$x \neq 0$$

$$= \frac{\frac{x^2 + 1 + x}{x}}{\frac{x^2 + 1 + 2x}{x}}$$

$$= \frac{x^2 + x + 1}{x} \times \frac{\cancel{x}}{x^2 + 1 + 2x}$$

$$= \frac{\cancel{x}^2 + x + 1}{(x+1)^2}$$

$$x \neq -1$$

$$D_{g \circ f} = \{x \mid x \neq 0 \text{ en } x \neq -1\}$$